Econ 381: Introduction to Econometrics
Handout 9: Nonlinear Regression Functions

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Fall 2026

Does the same class-size reduction predict the same gain?

Handout 8 used multiple regression to compare districts while holding their other characteristics fixed. With a single slope on the student–teacher ratio (STR), a drop of two students per teacher predicts the same score gain at any starting ratio.

The superintendent's next questions

Does reducing STR from 20 to 18 predict the same score gain as reducing it from 22 to 20? Does the gain depend on the share of English learners?

A **curve** lets the relationship change with the starting STR. An **interaction** lets the relationship vary with another district characteristic.

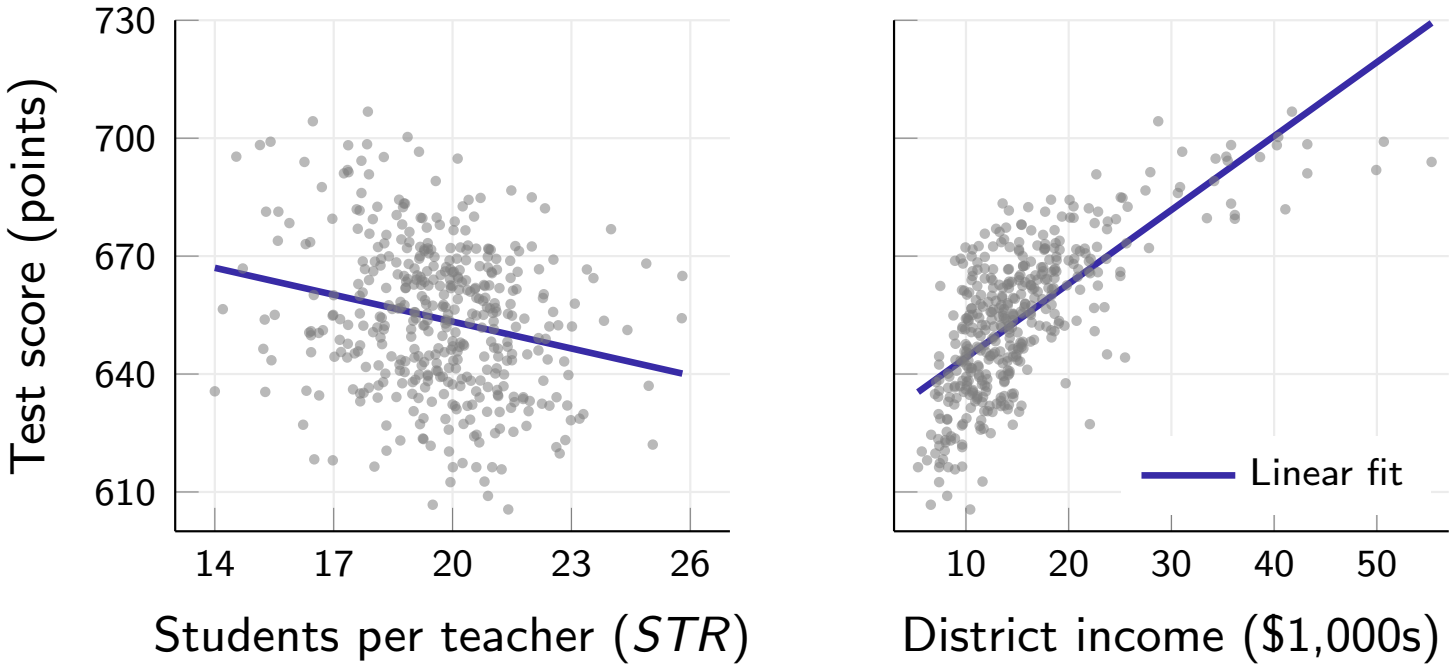
Roadmap

1. **8.1: A curve instead of a line.** Use district income to see why the same dollar increase can predict different score gains.
2. **8.2: Powers and logs.** Choose a functional form and explain the result in points or percentages.
3. **8.3: Interactions.** Let the relationship vary across groups or with another continuous characteristic.
4. **8.4: Back to class size.** Compare what a line and a curve predict for two equal cuts in STR, holding the same controls fixed.

Section 8.1

A general strategy for modeling nonlinear relationships

A straight line can miss a pattern in the data



Each dot is a district. In the right panel, scores rise quickly at low income levels and then level off. The fitted line predicts the same score gain per additional $1,000 at every income level.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Step 1. Give the regression room to bend

Let X_i be average district income in thousands of dollars per person. Add its square as a second regressor.

$$\text{TestScore}_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i.$$

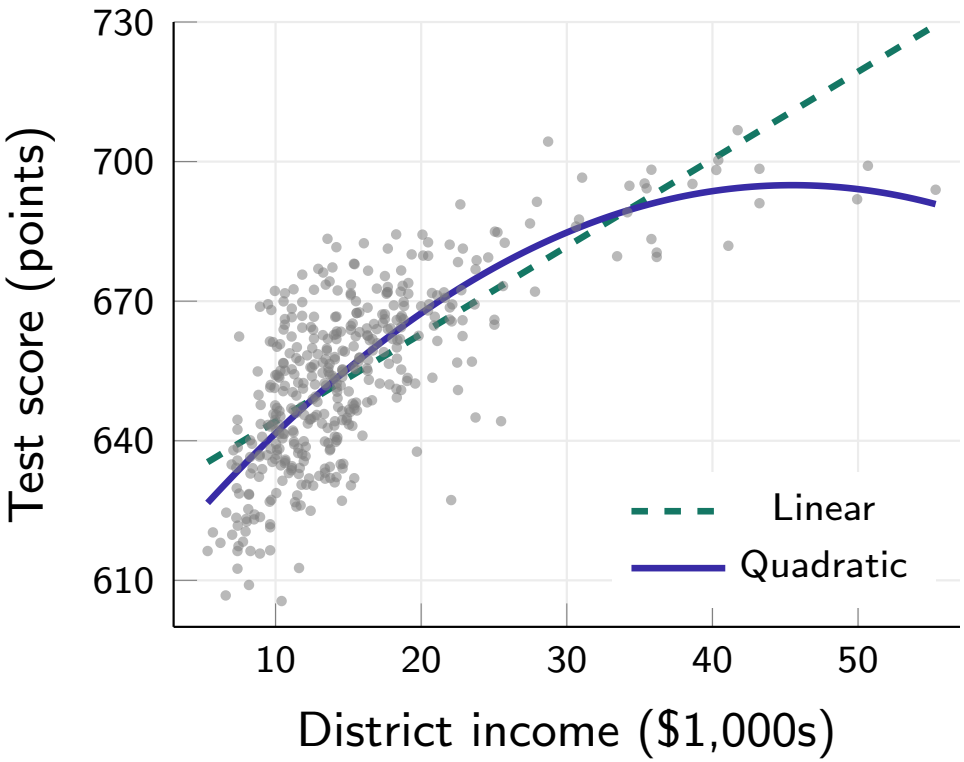
A change in income changes **both** X and X^2. The same dollar increase can move fitted scores by different amounts.

```
use caschool.dta, clear
gen avginc2 = avginc^2
regress testscr avginc avginc2, robust
```

Estimation is still OLS

Once we observe income, we can calculate both regressors. Stata estimates their coefficients by multiple regression.

Step 2. Check for curvature and plot the fit



Source: Stock and Watson, `caschool.dta`, 420 California school districts.

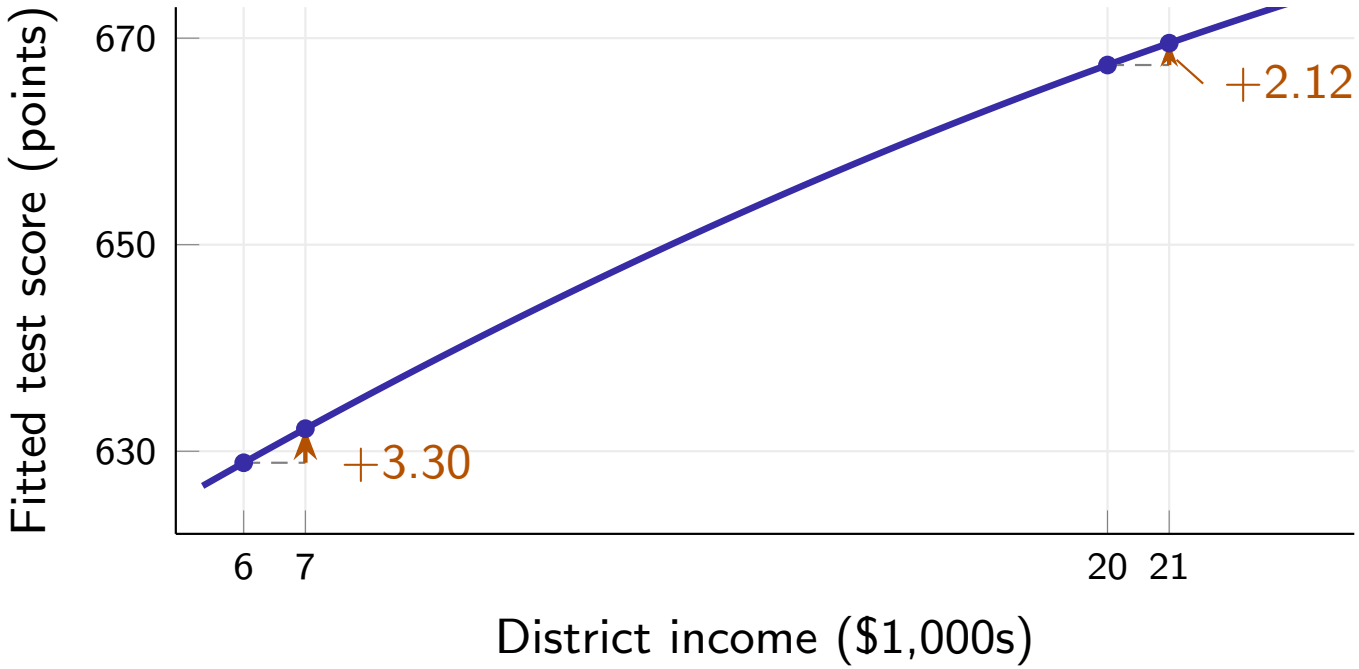
Is a line enough?

Under $H_0: \beta_2 = 0$, the quadratic reduces to a line.

$$\widehat{\beta}_2 = -0.04231,$$
$$\text{Robust SE} = 0.00478,$$
$$t = -8.85.$$

We reject the straight-line restriction at 5%. The fitted curve flattens as income rises, consistent with smaller gains from the same dollar increase.

Step 3. Compare predictions at two income levels



Both increases are \$1,000. The fitted score rises more from \$6,000 to \$7,000 than from \$20,000 to \$21,000.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Step 4. Use `lincom` for the prediction difference

When income goes from \$6,000 to \$7,000, X rises by 1 and X^2 rises by $49 - 36 = 13$. The intercept cancels when we subtract.

$$\widehat{\text{TestScore}}(7) - \widehat{\text{TestScore}}(6) = 1 \cdot \widehat{\beta}_1 + 13 \cdot \widehat{\beta}_2.$$

```
* After the quadratic regression
lincom avginc + 13*avginc2
```

Estimate	Robust SE	t	95% CI
3.30	0.21	15.90	[2.89, 3.71]

The estimated increase is 3.30 test-score points. Stata also tests whether the population prediction difference is zero.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Section 8.2

Nonlinear functions of one variable

A cubic adds one more power of income

A quadratic has constant curvature. A cubic gives the fit more flexibility by adding X^3 alongside X and X^2.

$$\text{TestScore}_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + u_i.$$

Use the joint F-test from Handout 8 to ask whether a line is enough: $H_0: \beta_2 = \beta_3 = 0$.

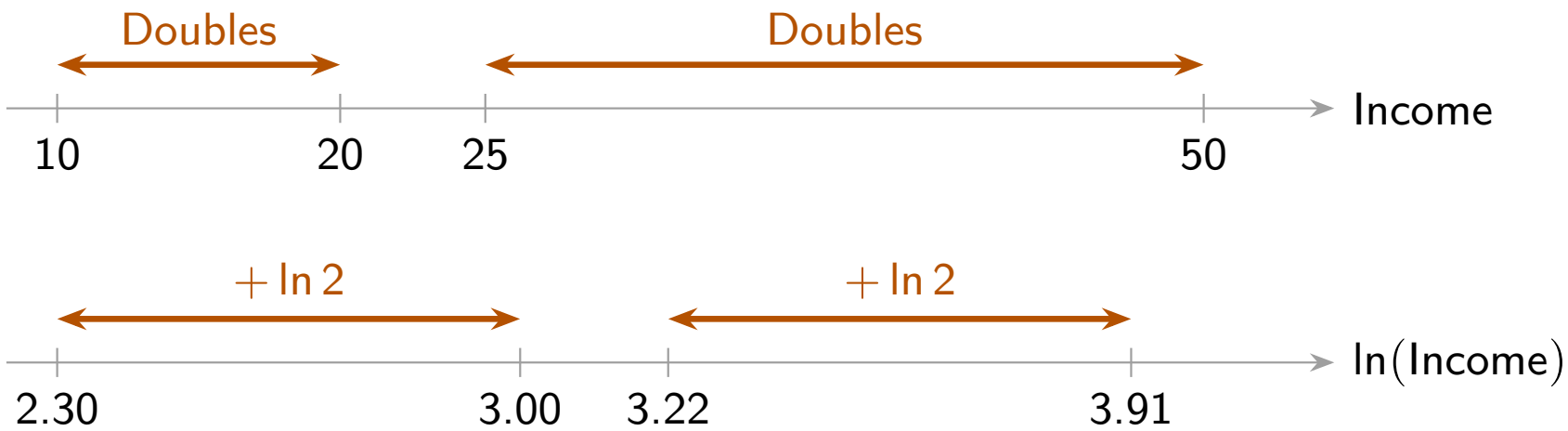
```
gen avginc3 = avginc^3
regress testscr avginc avginc2 avginc3, robust
test avginc2 avginc3
```

$F(2, 416) = 37.69$, with $p < 0.001$. The data favor allowing curvature.

To ask whether a quadratic is enough, test just $H_0: \beta_3 = 0$ in the cubic model. A log gives us another way to model curvature.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Logs measure changes in relative terms



Going from 10 to 20 and from 25 to 50 both double income. The dollar increases differ, but both add $\ln 2$ to log income.

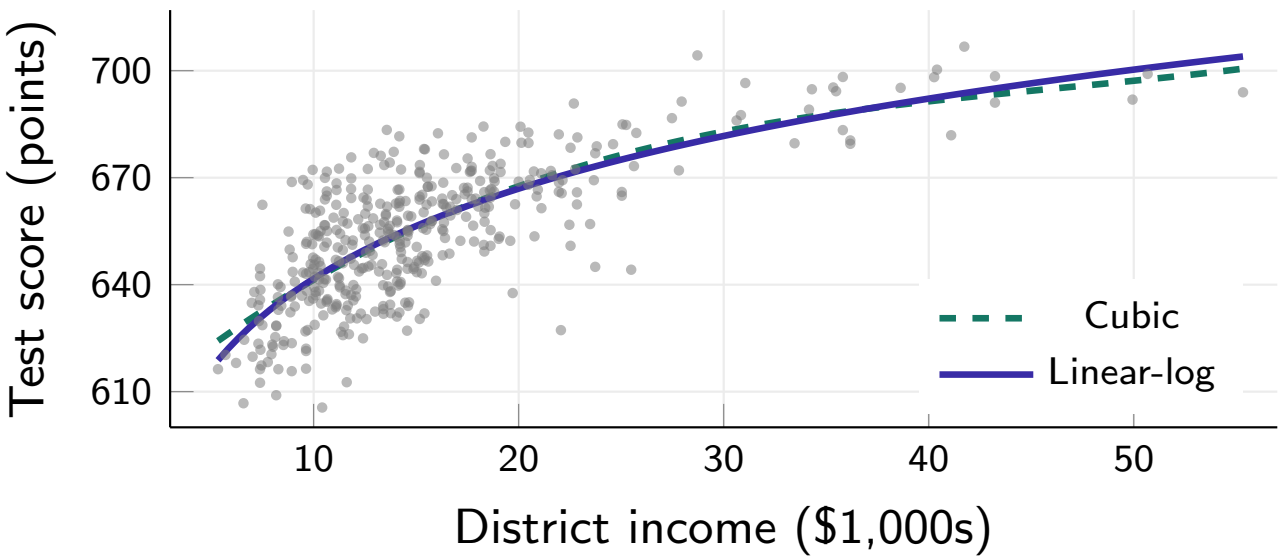
The shortcut for small changes

A 1% increase multiplies a positive value by 1.01, so its log rises by $\ln(1.01) \approx 0.01$.

Linear-log. Income in percent, scores in points

$$\text{TestScore}_i = \beta_0 + \beta_1 \ln(\textit{Income}_i) + u_i, \qquad \widehat{\beta}_1 = 36.42.$$

1. A 1% increase in income raises $\ln(\textit{Income})$ by about 0.01.
2. The fitted score rises by $36.42 \times 0.01 \approx 0.36$ points.



Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Log-linear. One more year gives a percent change

Illustration. Suppose we regress log hourly earnings on years of work experience.

$$\ln(Earnings_i) = \beta_0 + \beta_1 Experience_i + u_i.$$

The regression slope is $\widehat{\beta}_1 = 0.04$.

1. One more year raises fitted log earnings by 0.04.
2. A log change of 0.04 corresponds to about a 4% increase in earnings.

Read the units on both sides

The change in experience is measured in years. The fitted change in earnings is about 4% of the starting level. That is about \$0.80 an hour at \$20 and \$1.60 an hour at \$40.

Log-log. The slope is an elasticity

$$\ln(\text{TestScore}_i) = \beta_0 + \beta_1 \ln(\mathit{Income}_i) + u_i, \qquad \widehat{\beta}_1 = 0.0554.$$

1. A 1% increase in income changes its log by about 0.01.
2. The fitted log score rises by about 0.0554×0.01.
3. In percent terms, the fitted score rises by about 0.0554%.

Elasticity

In a log-log regression, β_1 gives the approximate percent change in Y associated with a 1% increase in X.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

How to read a log specification

Fitted outcome	Change in original X	Approximate change in original Y
$\widehat{Y} = \widehat{\beta}_0 + \widehat{\beta}_1 \ln(X)$	X increases by 1%	$0.01\widehat{\beta}_1$ units
$\widehat{\ln Y} = \widehat{\beta}_0 + \widehat{\beta}_1 X$	X increases by 1 unit	$100\widehat{\beta}_1\%$
$\widehat{\ln Y} = \widehat{\beta}_0 + \widehat{\beta}_1 \ln(X)$	X increases by 1%	$\widehat{\beta}_1\%$

When the outcome is logged, the percentages describe changes in the fitted outcome after converting it back to its original units.

Choose logs when percentages answer the question you care about. Compare R^2 across models only when they use the same outcome scale and sample.

In Stata, generate the log and run OLS

```
gen loginc = ln(avginc)
gen logscore = ln(testscr)

* Linear-log
regress testscr loginc, robust

* Log-linear
regress logscore avginc, robust

* Log-log
regress logscore loginc, robust
```

Here avginc and testscr are both positive in the California data.

Section 8.3

Interactions between independent variables

Can the relationship differ across districts?

Suppose two districts have the same student–teacher ratio, but one has a much larger share of English learners. We might see a score gap at the same ratio. We might also see a different score change when the ratio falls.

Two questions about two lines

- At a given STR, how far apart are the fitted scores?
- For the same drop in STR, how much does each fitted score rise?

A group indicator allows different intercepts. Multiplying the indicator by STR also allows different slopes. The product is an **interaction**.

Two binary variables give four group means

Let Y be the test score. Define $HiSTR = 1$ for $STR \geq 20$ and $HiEL = 1$ if at least 10% of students are English learners. Each indicator is zero below its cutoff.

$$Y_i = \beta_0 + \beta_1 HiSTR_i + \beta_2 HiEL_i + \beta_3 (HiSTR_i \times HiEL_i) + u_i.$$

	Lower STR	Higher STR	Higher minus lower STR
$HiEL = 0$	β_0	$\beta_0 + \beta_1$	β_1
$HiEL = 1$	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_1 + \beta_3$

- β_0 is the mean score when both indicators are zero.
- β_1 is the high-minus-low STR gap when $HiEL = 0$.
- β_2 is the mean for $HiEL = 1$ minus the mean for $HiEL = 0$ among districts with $STR < 20$.
- β_3 is the high-minus-low STR gap for $HiEL = 1$ minus the same gap for $HiEL = 0$.

California example. Estimate the four group means

```
gen histr = (str >= 20)
gen hiel = (el_pct >= 10)
regress testscr i.histr##i.hiel, robust
```

Here `el_pct` is *PctEL*. The prefix `i.` marks an indicator as categorical. `##` includes both indicators and their interaction.

Stata term	Coefficient	Estimate	Robust SE	p-value
`1.histr`	$\widehat{\beta}_1$	-1.908	1.932	0.324
`1.hiel`	$\widehat{\beta}_2$	-18.163	2.346	< 0.001
`1.histr#1.hiel`	$\widehat{\beta}_3$	-3.494	3.121	0.264
`_cons`	$\widehat{\beta}_0$	664.143	1.388	< 0.001

The interaction row tests whether the high-minus-low *STR* score gap is the same in both *HiEL* groups. With $p = 0.264$, we do not reject equal gaps at the 5% level.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

From Stata's coefficients to the four group averages

Start with the intercept, 664.143. Add the coefficient on each indicator that equals 1. Add the interaction coefficient when both equal 1.

	Lower *STR*	Higher *STR*	High-minus-low gap
$HiEL = 0$	$\widehat{\beta}_0$	$\widehat{\beta}_0 + \widehat{\beta}_1$	$\widehat{\beta}_1$
	664.14	662.24	−1.91
$HiEL = 1$	$\widehat{\beta}_0 + \widehat{\beta}_2$	$\widehat{\beta}_0 + \widehat{\beta}_1 + \widehat{\beta}_2 + \widehat{\beta}_3$	$\widehat{\beta}_1 + \widehat{\beta}_3$
	645.98	640.58	−5.40

When both indicators equal 1, the fitted mean is $664.143 - 1.908 - 18.163 - 3.494 = 640.578$ points. The high-minus-low *STR* score gap is about $|\widehat{\beta}_3| = 3.49$ points less in the $HiEL = 1$ group.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

A binary-by-continuous interaction gives two slopes

Now keep $X = STR$ continuous and interact it with the indicator $D = HiEL$.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i D_i) + u_i.$$

Group	Fitted line	Slope on X
$D = 0$	$\widehat{\beta}_0 + \widehat{\beta}_1 X$	$\widehat{\beta}_1$
$D = 1$	$(\widehat{\beta}_0 + \widehat{\beta}_2) + (\widehat{\beta}_1 + \widehat{\beta}_3) X$	$\widehat{\beta}_1 + \widehat{\beta}_3$

$\widehat{\beta}_2$ is the fitted score for $D = 1$ minus the fitted score for $D = 0$ at $X = 0$. At any X, the score gap is $\widehat{\beta}_2 + \widehat{\beta}_3 X$.

$\widehat{\beta}_3$ is the slope for $D = 1$ minus the slope for $D = 0$. It tells us how much the fitted score change per additional student per teacher differs between the two groups.

From Stata's coefficients to the two fitted lines

```
regress testscr c.str##i.hiel, robust
```

Stata term	Estimate	Robust SE	p-value
`str`	-0.968	0.589	0.101
`1.hiel`	5.639	19.515	0.773
`1.hiel#c.str`	-1.277	0.967	0.187
`_cons`	682.246	11.868	< 0.001

For $HiEL = 0$, use the constant and the slope on `str`. For $HiEL = 1$, add `1.hiel` to the intercept and add `1.hiel#c.str` to the slope.

$$HiEL = 0: \quad \widehat{\text{TestScore}} = 682.246 - 0.968\, STR,$$

$$HiEL = 1: \quad \widehat{\text{TestScore}} = (682.246 + 5.639) + (-0.968 - 1.277)\, STR$$
$$= 687.885 - 2.245\, STR.$$

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

In Stata, test the slope and the gap separately

After the interaction regression, we can use these two commands to answer different questions:

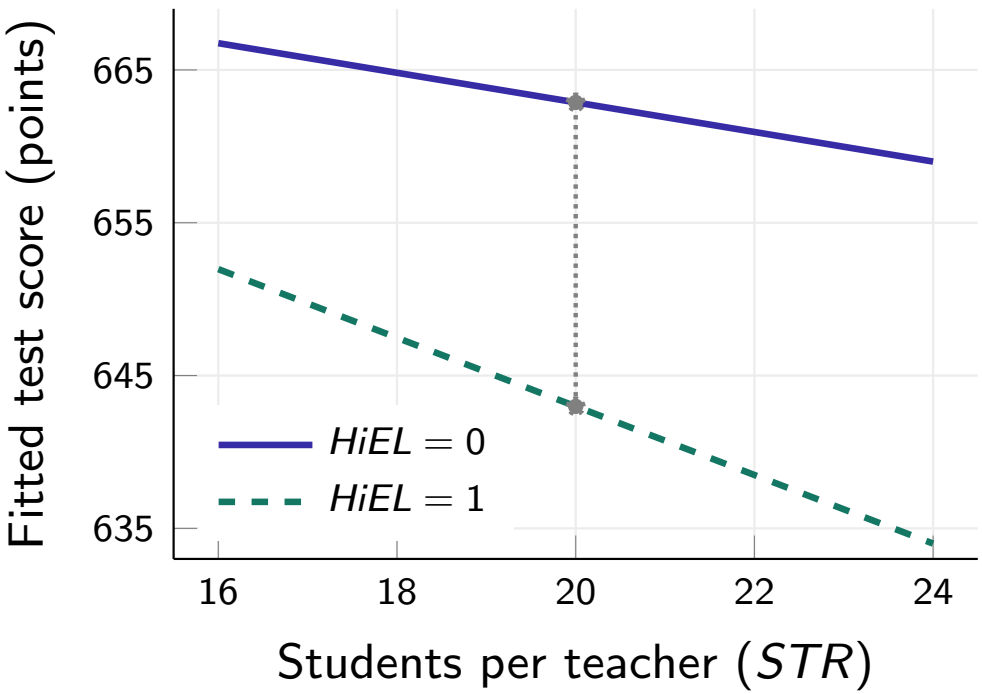
```
lincom str + 1.hiel#c.str
test 1.hiel#c.str = 0
```

- `lincom` tests whether the slope for the group $HiEL = 1$ is zero. That slope is -2.25 points per additional STR ($p = 0.004$).
- `test` asks whether the two slopes are equal. The estimated slope difference is -1.28 points per additional STR ($p = 0.187$).

We reject a zero slope for the group with at least 10% English learners at the 5% level. We do not reject equal slopes across these two groups.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Do smaller classes predict larger gains for one group?



At $STR = 20$, the dotted line marks the gap between the fitted scores of 663 ($HiEL = 0$) and 643 ($HiEL = 1$).

For a cut in STR from 20 to 19, the fitted score rises by

- 0.97 points for $HiEL = 0$,
- 2.25 points for $HiEL = 1$.

The gain is 1.28 points larger in the second group.

Why might districts with more English learners benefit more?

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Let the slope vary with the share of English learners

With a 10% cutoff on *PctEL*, districts at 9% and 10% can have different fitted slopes. Using *PctEL* directly lets the slope change gradually as the share of English learners rises.

Let $X = STR$ and $Z = PctEL$, measured on a 0–100 scale.

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i + u_i.$$

- Holding Z fixed, one more student per teacher changes the fitted score by $\beta_1 + \beta_3 Z$.
- If the English-learner share rises by 10 percentage points, that fitted slope changes by $10\beta_3$.

How much does the fitted slope change with *PctEL*?

```
regress testscr c.str##c.el_pct, robust
```

Stata term	Estimate	Robust SE	p-value
`str`	-1.1170	0.5875	0.058
`el_pct`	-0.6729	0.3741	0.073
`c.str#c.el_pct`	0.00116	0.01854	0.950
`_cons`	686.3385	11.7594	< 0.001

Fitted slope on $STR = -1.1170 + 0.00116\, PctEL$.

Share of English learners	0%	10%	30%
Fitted slope on *STR*	-1.12	-1.11	-1.08

Moving from 0% to 30% adds about 0.04 points to the fitted slope. The interaction's $p = 0.950$ gives little evidence that the slope varies with *PctEL* in this model.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Section 8.4

Nonlinear effects of the student–teacher ratio on California test scores

Does the starting class size matter?

Compare two districts. One lowers its student–teacher ratio from 22 to 20. The other lowers it from 20 to 18. Both cuts are two students per teacher. Do we expect the same score gain?

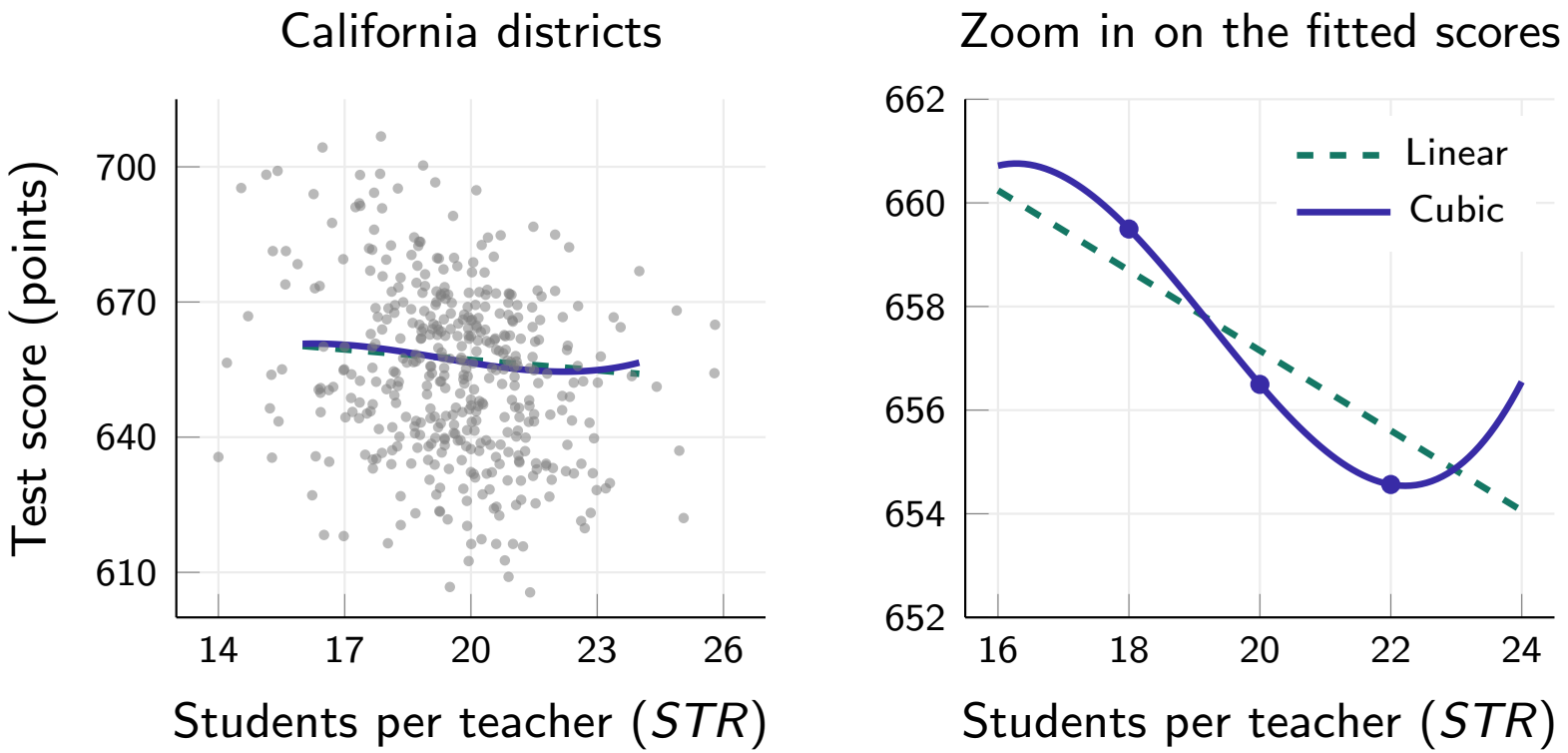
A cut might matter more once classes are small enough for teachers to work with students in small groups. We can let the data show whether the fitted gains vary with the starting ratio.

Fit two models to the California districts

- **Linear.** Include STR. A two-point cut predicts the same gain anywhere on the line.
- **Cubic.** Include STR, STR^2, and STR^3. The curve can be steeper over one range than another.

Both models include $HiEL$, the percentage eligible for subsidized lunch, and $\ln(Income)$. Using the same controls lets us compare what changes when we allow the STR relationship to be nonlinear.

The curve is steeper between 18 and 20



Dots are observed district scores. Both fits hold $HiEL = 0$, lunch eligibility at 45%, and district income at \$15,000. For each cut, move left along the same fit. On the cubic, the vertical rise is larger from 20 to 18.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

In Stata, fit the line and the cubic

```
gen strsq = str^2
gen strcu = str^3
regress testscr str hiel meal_pct loginc, robust
regress testscr str strsq strcu hiel meal_pct loginc, robust
```

Regressor	Linear	Cubic
STR	−0.772 (0.256)	64.340 (24.861)
STR^2		−3.424 (1.250)
STR^3		0.05929 (0.02076)
HiEL	−5.791 (1.027)	−5.474 (1.034)
Lunch eligibility	−0.417 (0.028)	−0.420 (0.029)
ln(*Income*)	11.820 (1.775)	11.748 (1.771)
Constant	659.356 (8.421)	252.046 (163.634)
Districts	420	420

Robust standard errors in parentheses.

In the cubic, a cut in *STR* changes all three terms at once. To find the gain, compare two fitted scores.

In Stata, calculate the gains and test the curvature

In either model, subtract the fitted score at $STR = 22$ (or 20) from the fitted score at $STR = 20$ (or 18). The controls cancel because they stay fixed.

```
regress testscr str hiel meal_pct loginc, robust
lincom -2*str

regress testscr str strsq strcu hiel meal_pct loginc, robust
lincom -2*str - 84*strsq - 2648*strcu  // 22 to 20
lincom -2*str - 76*strsq - 2168*strcu  // 20 to 18
test strsq strcu
lincom 8*strsq + 480*strcu             // second gain minus first
```

For $20 \to 18$, the weights on coefficients are $18 - 20 = -2$ for β_{STR}, $18^2 - 20^2 = -76$ for β_{STR^2}, and $18^3 - 20^3 = -2168$ for β_{STR^3}.

Subtract the two sets of weights to test whether the gains are equal between $22 \to 20$ and $20 \to 18$. The STR weight is -2 for both cuts, so it drops out.

The joint test that the coefficients on STR^2 and STR^3 are both zero gives $F(2, 413) = 6.17$ and $p = 0.0023$.

Adding curvature changes the predicted gains

Holding each district's other characteristics fixed gives the following fitted score gains.

Cut in *STR*	Linear gain	Cubic gain	95% CI for cubic gain
$22 \to 20$	$+1.54$	$+1.93$	$[0.61, 3.25]$
$20 \to 18$	$+1.54$	$+3.00$	$[1.64, 4.36]$

All gains are in test-score points. With a line, both cuts look the same. With a cubic, the cut from 20 to 18 has a fitted gain about one point larger than the cut from 22 to 20.

What does the evidence say?

Equal gains means the score increase from 20 to 18 equals the increase from 22 to 20, with controls held fixed. The cubic estimates a difference of 1.07 points, with a 95% CI of $[-0.01, 2.14]$. Test of equal gains for these two cuts gives $p = 0.052$, despite rejecting the straight line ($p = 0.0023$ for $H_0 : \beta_{STR^2} = \beta_{STR^3} = 0$).

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

In Stata, allow a different curve for each group

Let $X = STR$, $H = HiEL$, L be lunch eligibility in percent, and I be district income. The fitted score is

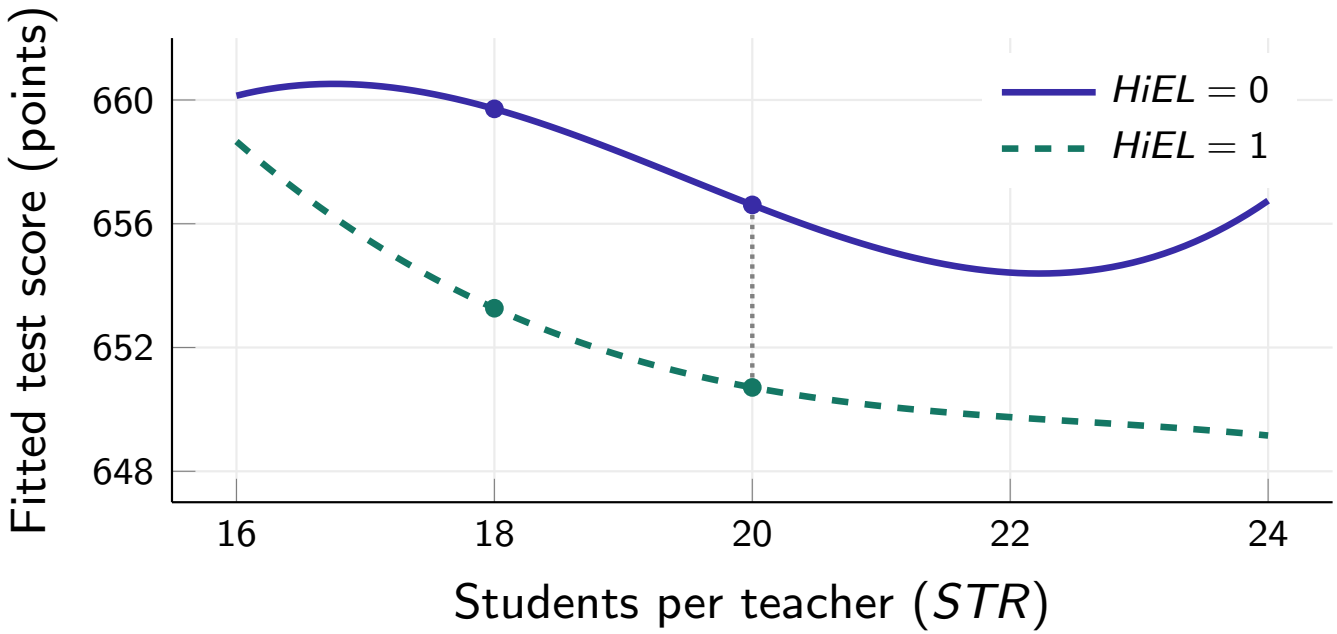
$$\widehat{\text{TestScore}} = \widehat{\beta}_0 + \widehat{\beta}_1 X + \widehat{\beta}_2 X^2 + \widehat{\beta}_3 X^3 + H(\widehat{\delta}_0 + \widehat{\delta}_1 X + \widehat{\delta}_2 X^2 + \widehat{\delta}_3 X^3) + \widehat{\gamma}_1 L + \widehat{\gamma}_2 \ln(I).$$

```
gen strh = str*hiel
gen strsqh = strsq*hiel
gen strcuh = strcu*hiel
regress testscr str strsq strcu hiel strh strsqh strcuh ///
    meal_pct loginc, robust
```

For $HiEL = 0$, the terms multiplied by H are zero. For $HiEL = 1$, those terms can change both the level and the slope of the fitted curve.

The curve can also differ across groups

Hold district income at \$15,000 and lunch eligibility at 45% for both groups. Compare scores at the same STR by looking across curves. Compare score changes by moving along each curve.



The dotted line marks the score gap at $STR = 20$. The dots at 18 and 20 show the endpoints of fitted scores.

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Read the levels, slopes, and gains from the curves

- **Level.** At $STR = 20$, the fitted score is about 5.90 points lower for districts with at least 10% English learners.
- **Slope.** At $STR = 20$, the curve for $HiEL = 0$ falls more steeply. Its slope there is -1.62 points per additional student per teacher, compared with -0.78 for $HiEL = 1$. Each slope varies with STR.

Group	Score at $STR = 20$	Score at $STR = 18$	Fitted gain
$HiEL = 0$	656.61	659.71	$+3.10$
$HiEL = 1$	650.71	653.27	$+2.56$

Gain from 20 to 18. The fitted gain is 0.54 points larger for $HiEL = 0$. The data give little evidence of a difference for this particular cut in class size ($p = 0.698$).

Source: Stock and Watson, `caschool.dta`, 420 California school districts.

Summary. What does a nonlinear model let us ask?

- **Does the starting value matter?** Powers such as X^2 and X^3 let the same change in X predict different changes in Y.
- **Which units make sense?** Logs help express small changes in percent. With both variables logged, the slope is an elasticity.
- **Does the relationship vary by group?** Interactions let a slope or a whole curve depend on another variable.
- **What is the comparison of interest?** Hold the controls fixed and subtract the two fitted values. Use `lincom` to report uncertainty for the comparison and `test` to assess several coefficients jointly.